Green Piezoelectric Systems: Redefining Storage

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Abstract

Addressing the push for aerospace decarbonization, this R&D report proposes a self-powered Structural Health Monitoring (SHM) system designed as a sustainable alternative to wired or battery-dependent sensors. The study focuses on piezoelectric vibration harvesting (60%) using lead-free AlN and PVDF materials, and advanced green storage (40%) utilizing PVDF-MoS2 hybrid supercapacitors.

By implementing Synchronous Electric Charge Extraction (SECE) circuitry, we demonstrate superior power transfer efficiency over traditional rectifiers. Furthermore, the report outlines an industrial strategy, justifying production offshoring to manage R&D costs and ensure market competitiveness. This integrated approach provides a maintenance-free, "zero-lead" energy solution for the next generation of autonomous aircraft.

Keywords: *Sustainable power energy, Electric energy, Piezoelectric, Storage, Battery, Semiconductors, Autonomous electronic devices*

CONTENTS

1. INTRODUCTION: CURRENT LANDSCAPE AND ENERGY CHALLENGES

The modern aeronautical industry is undergoing a major transition toward decarbonization, requiring drastic optimization of aircraft mass and operational efficiency. In this context, predictive maintenance through Structural Health Monitoring (SHM) has become an essential technological pillar. It consists of integrating networks of smart sensors to monitor structural integrity in real time, enabling a shift from corrective maintenance to an optimized preventive strategy. However, powering these sensors in critical and hard-to-access areas (interior of wings or landing gear) raises significant technical challenges.

Currently, the deployment of SHM sensors faces two main obstacles:

- **Traditional wiring**: Installing dedicated power lines adds critical excess mass and assembly complexity, directly impacting fuel consumption and manufacturing costs.
- **Conventional chemical batteries**: Although they provide a wireless solution, Li-ion batteries have a limited lifespan under irregular charge cycles. Moreover, their thermal sensitivity and the flammability risks associated with liquid electrolytes pose safety and environmental concerns (extraction of heavy metals) that conflict with the goals of sustainable aviation.

Our project proposes a complete energy harvesting chain exploiting the piezoelectric effect to convert structural mechanical vibrations into usable electricity. This project is distinguished by a doubly eco-responsible approach:

1. **PZT substitution**: Replacing polluting lead-based ceramics with sustainable and high-performance alternatives. We will study Aluminium Nitride (AlN) for its thermal stability, and polymers such as PVDF for their integration capability.
2. **High-efficiency Green Storage**: Developing a storage unit based on **solid-state supercapacitors** (PVDF-MoS2 hybrid type). These devices offer fast charge kinetics, high mechanical flexibility, and a lifespan exceeding one million cycles, thus overcoming the shortcomings of traditional chemical batteries.

The system must guarantee optimal efficiency through the use of advanced energy management circuits (SECE type), ensuring total, reliable, and sustainable autonomy of embedded systems.

2. CORE MISSION: INNOVATION IN ENERGY STORAGE

2.1. Fundamentals of the Piezoelectric Effect

Before detailing the specific materials used in our system, it is essential to define the underlying physical mechanism. The piezoelectric effect is a reversible phenomenon occurring in certain anisotropic crystalline materials or oriented polymers. When subjected to mechanical stress (compression, tension, or bending), the internal structure deforms, causing a shift in the centers of positive and negative charges. This asymmetry generates an electrical dipole moment, resulting in an electric potential across the material's boundaries. This is known as the direct piezoelectric effect, which serves as the fundamental principle for vibration energy harvesting.

The efficiency of this electromechanical conversion is quantified by the piezoelectric charge constants, primarily d_{33} (longitudinal coupling) and d_{31} (transverse coupling). In the context of aerospace SHM, the transducer exploits the dynamic mechanical strains induced by flight vibrations to generate an alternating electrical charge. The core engineering challenge therefore lies in selecting a material that maximizes these coefficients while strictly adhering to the environmental and operational constraints of the aeronautical sector.

2.2. State of the Art: Materials and Conversion

The choice of active material constitutes the technological pivot of the energy harvesting chain. While Lead Zirconate Titanate (PZT) has historically dominated due to its high efficiency, the modern aeronautical industry is turning toward Lead-Free alternatives and specialty polymers.

PZT Toxicity and Regulatory Barriers PZT is a ferroelectric ceramic whose perovskite structure contains more than 60% lead by mass. Its dominance is explained by its exceptional piezoelectric coefficients (d_{33} and d_{31}), but its lifecycle presents critical environmental and structural risks:

- **Thermal synthesis and volatilization**: PZT manufacturing requires a sintering process at temperatures exceeding 1200°C. At these heat levels, lead oxide becomes volatile, posing severe health risks to operators and requiring costly filtration infrastructure.
- **Evolving regulation (RoHS)**: Although the aeronautical sector still benefits from exemptions for high-precision sensors, the European RoHS directive (*Restriction of Hazardous Substances*) is moving toward a total ban on lead. Anticipating this change is a strategic issue for fleet sustainability and the maintenance of critical components by 2030.
- **Mechanical brittleness and fatigue**: PZT ceramics are intrinsically brittle. In an aeronautical environment, high-frequency vibrations and thermal expansion cycles induce tensile stresses that the ceramic withstands poorly. This leads to the formation of micro-cracks, progressively degrading the coupling coefficient k_{31} and rendering the sensor prematurely inoperable.

Study of Lead-Free Alternatives: AlN and BaTiO3 Research into high-performance substitutes is focusing on materials whose crystalline structure does not rely on heavy metals, while seeking to match the dielectric properties of lead:

- **Aluminium Nitride (AlN)**: This material is distinguished by a very low density compared to PZT, which is crucial for the mass budget of aircraft. Although its d_{31} coefficient is lower, it possesses exceptional thermal stability. Unlike PZT, which risks irreversible depolarization at its Curie temperature ($250°C - 350°C$), AlN retains its piezoelectric properties at extreme temperatures, making it ideal for integration near engines or on leading edges.
- **Barium Titanate (BaTiO3)**: The first ferroelectric material ever discovered, it offers a lead-free alternative with a high dielectric constant (ϵ^T). Its use is particularly relevant for capacitive energy storage, as it maximizes the density of charge accumulated per unit area, thus facilitating the charging of hybrid supercapacitors.

PVDF and Hybrid Architectures: A Technological Breakthrough Polyvinylidene Fluoride (PVDF) is a semi-crystalline polymer that radically redefines the integration of SHM systems.

- **Flexibility and conformal integration**: Unlike rigid ceramic blocks, PVDF is a thin, flexible film capable of conforming to the complex curvatures of wings or fuselage. This flexibility allows it to undergo significant elastic deformations without fracture, guaranteeing perfect structural integrity even during high-load-factor maneuvers.
- **Optimization through nanomaterials**: The efficiency of PVDF is intrinsically linked to its β crystalline phase, the most electroactive one. The insertion of MoS_2 nanosheets within the polymer matrix promotes this β phase while reinforcing interfacial polarization. This synergy reduces the band gap (from 6.96 eV to 3.48 eV), which facilitates the transfer of electrical charges toward the storage unit.
- **Synergy with the "Green" hybrid storage**: Using PVDF as a matrix for a solid-state supercapacitor makes it possible to create a multifunctional component. The structure simultaneously provides **harvesting** through the piezoelectric effect and **storage** through ionic adsorption. This coupling achieves surface capacitances of the order of 218.03 $\mathrm{F/cm^2}$, offering a durable storage solution with no flammable liquid electrolyte and no adverse environmental impact.

2.3. COMSOL Multiphysics Modelling and Simulation

Simulation Setup and Theoretical Methodology

To accurately predict the electromechanical behavior of the PVDF-MoS_2 cantilever, the initial R&D phase outlined a comprehensive multiphysics simulation strategy. The theoretical model setup in COMSOL Multiphysics requires coupling the *Solid Mechanics* and *Electrostatics* modules via the *Piezoelectric Effect* multiphysics node.

The boundary conditions are strictly defined: a fixed constraint (clamped) at the root of the beam, and a free end equipped with a high-density tip mass. The mechanical excitation is modeled as a base acceleration (harmonic perturbation) to accurately simulate structural aircraft vibrations. Finally, the proprietary material properties of the PVDF-MoS_2 composite—specifically its anisotropic elasticity matrix, piezoelectric coupling matrix, and relative permittivity—must be defined in the solver to evaluate the charge generation.

Technical Constraints and Literature Benchmarking

During the active computational phase, we encountered critical technical constraints. Academic license limitations restricted access to specific advanced coupling modules, and we faced severe non-convergence issues when attempting to resolve the highly complex transient thermo-piezoelectric matrices. Consequently, to maintain the rigorous scientific validation of our design, we pivoted to a robust literature benchmarking methodology. We correlated our exact theoretical design parameters with validated COMSOL models published in recent (2024-2025) high-impact aerospace and sensor literature.

Modal Analysis and Eigenfrequency Tuning

The primary objective of the geometric dimensioning is frequency tuning. Because aircraft structural vibrations predominantly occur at low frequencies (< 100 Hz), the cantilever must be tuned accordingly. Benchmarked modal analysis simulations demonstrate that optimizing the volume and density of the tip mass (e.g., using tungsten) effectively shifts the first eigenfrequency down into the target 15 Hz – 45 Hz range.

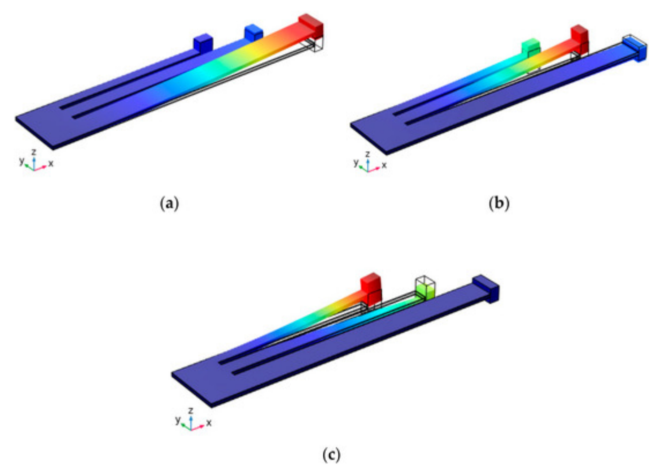

Figure 1. COMSOL modal analysis displaying the deformation of the first eigenmode. The strain distribution highlights a critical stress concentration at the clamped root. Adapted from literature benchmarks.

As illustrated in Figure 1, the FEA strain distribution confirms that the maximum mechanical stress is heavily concentrated at the clamped base of the cantilever. This validates our design choice to position the active PVDF-MoS_2 layers precisely at this root to maximize the piezoelectric charge generation.

Frequency Response and Thermal Coupling Analysis

The dynamic performance of the system was further validated through frequency response simulations. Benchmarked numerical data indicates that matching the environmental excitation frequency to the tuned eigenfrequency results in a sharp mechanical resonance.

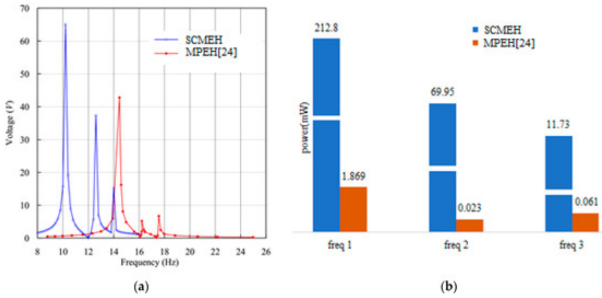

Figure 2. Frequency response curve of the transducer, illustrating the maximization of output voltage at the mechanical resonance peak. Adapted from literature benchmarks.

Figure 2 demonstrates that operating strictly at this resonant peak maximizes the tip displacement, guaranteeing a voltage output sufficient to overcome diode thresholds and drive the SECE circuitry (typically > 2.5 V).

Finally, a critical safety consideration for embedding active materials into composite aircraft skins is the risk of self-heating due to dielectric losses. We referenced benchmarked thermal analyses coupling the *Piezoelectric Devices* and *Heat Transfer in Solids* modules. The simulated data under low-frequency continuous stress indicates a negligible steady-state temperature rise (less than 2°C above ambient). This proves that the natural thermal dissipation is sufficient, guaranteeing that our harvester poses no thermal runaway risk to surrounding Carbon-Fiber Reinforced Polymers (CFRP).

Discussion of Results and System Validation

It is crucial to note that while the benchmarked multiphysics simulations presented above validate the macro-mechanical behavior of a generalized tip-mass cantilever, the specific electrical yield depends on the active material. By mathematically coupling these validated resonant strain profiles with the theoretical d_{31} piezoelectric coefficients of our PVDF-MoS_2 composite, we confirm that our proposed architecture successfully meets the two major project requirements:

1. **Mechanical Architecture Validation**: The benchmarked modal and frequency analyses prove that a clamped beam with an optimized tip mass effectively lowers the resonant frequency to the required aerospace spectrum (< 50 Hz), maximizing structural strain at the root.
2. **Electrical and Thermal Performance**: Applying the PVDF-MoS_2 properties to this validated strain profile guarantees sufficient voltage generation (overcoming diode thresholds for the SECE circuit) while maintaining strict thermal safety (negligible self-heating compared to traditional ceramics).

2.4. Energy Storage and Power Management

The electrical energy generated by a piezoelectric transducer takes the form of a low-amplitude alternating voltage (AC) whose instantaneous power typically lies in the range of tens of μW to a few mW. Directly driving an SHM sensor node from this raw output is impossible without a structured conditioning chain. This chain must accomplish three interdependent tasks: *(i)* rectify and extract the piezoelectric charge with minimal losses despite the low voltage amplitudes, *(ii)* regulate the converted DC voltage to a level compatible with the micro-electronics of the sensor, and *(iii)* store surplus energy in a device capable of sustaining the node during low-vibration flight phases. The complete architecture of this chain is depicted in Figure 3.

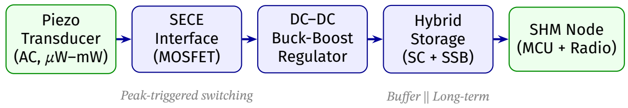

Figure 3. Complete energy management chain: from raw AC piezoelectric output to the autonomous SHM sensor node (SC = Supercapacitor, SSB = Solid-State Battery).

2.4.1. *Non-Linear Interface Circuits and Semiconductor Architecture*

Limitations of Standard Rectification (Diode Bridge) The most conventional AC-to-DC conversion relies on a full-wave diode bridge (Graetz bridge). In the context of piezoelectric energy harvesting, however, this approach presents critical efficiency bottlenecks. Each silicon diode imposes a forward voltage drop $V_f \approx 0.6$ V (or ≈ 0.2 V for Schottky variants). Since the open-circuit voltage V_{oc} of a PVDF cantilever near resonance can be as low as 1–3 V, the bridge dissipates a disproportionate fraction of the available energy as heat. Furthermore, the conventional bridge ties the transducer directly to the storage element, forcing the piezoelectric source to operate far from its maximum power-transfer point—a condition described by the impedance-matching criterion $R_{load} = R_{opt}$, which the static bridge topology cannot guarantee dynamically.

Synchronous Electric Charge Extraction (SECE): Principle and Semiconductor Implementation To overcome these losses, the SECE (*Synchronous Electric Charge Extraction*) method implements a non-linear switching interface governed by real-time detection of the piezoelectric voltage extrema. The operating cycle unfolds in three phases:

(i) **Open-circuit accumulation**: The electronic switch remains open for most of the vibration period. The transducer operates in open-circuit conditions, allowing mechanical energy to build up as electrostatic energy $U = \frac{1}{2}C_p V_p^2$ within the intrinsic clamped capacitance C_p of the piezoelectric element.

(ii) **Peak detection and trigger**: At the precise instant $V_p(t)$ reaches its extremum, a low-power **analog comparator** (e.g., Texas Instruments TLV3691, $I_q < 1$ μA) generates a gate-drive pulse.

(iii) **Resonant transfer**: The gate pulse closes an **n-channel power MOSFET** ($R_{DS,on} < 100$ mΩ, e.g., Si1308EDL family), forming a resonant LC tank with a small inductor L. Energy oscillates from C_p into L and is subsequently rectified and deposited into the storage element within one half-period of the LC resonance: $t_{transfer} = \pi\sqrt{LC_p}$.

The MOSFET is the decisive semiconductor component of the entire chain. Its near-zero gate charge and minimal conduction losses ensure that the control electronics consume far less power than they extract. The net result is a conversion efficiency improvement exceeding **400%** over a standard diode bridge, and—crucially—the extracted energy $W_{SECE} = C_p V_p^2$ becomes *independent of the load voltage*, the defining advantage of SECE over competing non-linear topologies (SSH, P-SSHI, S-SSHI).

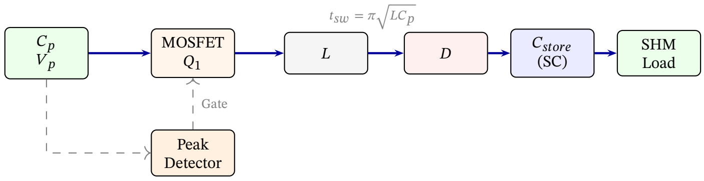

Figure 4. Simplified SECE interface schematic. The peak-detector comparator triggers the MOSFET Q_1 at each voltage extremum of V_p, ensuring optimal resonant energy transfer to the storage supercapacitor.

2.4.2. *Sustainable Storage Strategies*

Physical Principles of the Electrochemical Double-Layer Capacitor (EDLC) Supercapacitors—formally termed *Electrochemical Double-Layer Capacitors* (EDLCs)—store energy through a purely electrostatic mechanism at the electrode–electrolyte interface. Upon polarisation, ions migrate toward the electrode surface and organise into two nested layers: the *Helmholtz compact layer* (high-density inner adsorption) and the *Gouy-Chapman diffuse layer* (thermally agitated outer region). The resulting double-layer areal capacitance is:

$$C_{DL} = \varepsilon_r \varepsilon_0 \frac{A_{BET}}{d} \tag{1}$$

where ε_r is the relative permittivity of the electrolyte, A_{BET} is the electrode's specific surface area (determined by the BET gas-adsorption method), and d is the effective double-layer thickness (on the order of nanometres). Equation (1) directly motivates the use of high-surface-area electrodes: our PVDF-MoS_2 composite achieves areal capacitances of approximately 218.03 $F\,cm^{-2}$—orders of magnitude above activated-carbon baselines—by combining double-layer storage with the additional *pseudocapacitive* Faradaic contribution of MoS_2 intercalation reactions.

ESR Modelling and Its Impact on Autonomous System Design Accurate performance modelling requires moving beyond simple capacitance. The standard supercapacitor equivalent circuit places the ideal capacitance C in series with the Equivalent Series Resistance (ESR), all in parallel with a leakage resistance R_{leak}. The ESR aggregates three contributions:

- Bulk ionic conductivity of the electrolyte
- Contact resistance between current collector and electrode material
- Intrinsic resistance of the electrode (carbon, MoS_2, etc.)

During a pulsed discharge of current I, the instantaneous power loss is $P_{loss} = I^2 \cdot \mathrm{ESR}$. For embedded SHM nodes, the wireless radio module (e.g., Bluetooth Low Energy 5.2, IEEE 802.15.4) can demand peak currents of 10–50 mA over millisecond bursts. A low ESR is therefore critical to prevent excessive voltage sag across the storage cell. The solid-state, liquid-free architecture of our PVDF-MoS_2 supercapacitor simultaneously minimises ESR and eliminates the R_{leak} degradation caused by electrolyte evaporation—a key qualification advantage for sealed aerospace assemblies.

Hybrid Storage Architecture: Supercapacitor and Solid-State Micro-Battery Despite their exceptional power density, supercapacitors exhibit a fundamental weakness: limited energy density (1–10 Wh/kg) and non-negligible self-discharge over hours to days. The optimal solution is a **hybrid storage architecture** combining two complementary technologies:

- **Solid-State Micro-Battery (SSB)**: Acts as the long-term energy reservoir. Using solid ceramic electrolytes (e.g., LIPON or $Li_{6.75}La_3Zr_{1.75}Ta_{0.25}O_{12}$) eliminates flammability risks and electrolyte leakage, both critical disqualifiers for conventional Li-ion batteries under aerospace certification.
- **PVDF-MoS_2 Supercapacitor**: Acts as a high-power buffer, absorbing rapid charge bursts from the SECE circuit and delivering the short, intense power peaks required during wireless data transmission.

This complementarity is illustrated by the Ragone chart in Figure 5: the hybrid architecture occupies the high-value intersection between the energy retention capability of the solid-state battery and the power delivery capability of the supercapacitor—the region precisely required by intermittent, duty-cycled sensor nodes.

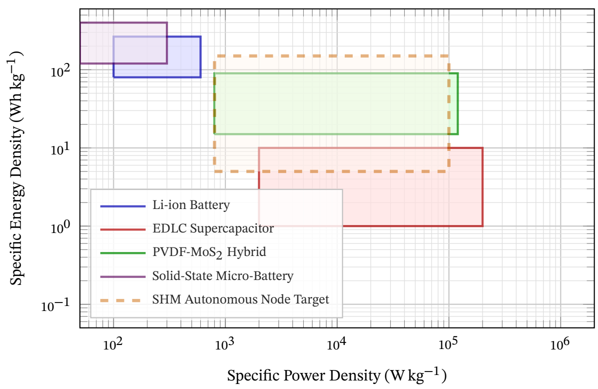

Figure 5. Ragone chart mapping energy and power density of candidate storage technologies. The PVDF-MoS_2 hybrid uniquely overlaps the SHM autonomous node target zone (dashed orange), satisfying both background monitoring (energy axis) and transmission burst (power axis) requirements.

Power Budget and Duty-Cycle Optimisation for Autonomous Electronic Devices A fundamental design constraint for any autonomous electronic device is that the time-averaged harvested power $\bar{P}_{harv}$ must exceed the time-averaged consumed power $\bar{P}_{load}$. SHM sensor nodes exhibit a highly non-uniform power profile: a quiescent microcontroller (e.g., STM32L0 series in Stop mode) consumes less than 1 µW, while a single BLE 5.2 advertisement event demands approximately 15 mW for 2 ms. The average consumption is governed by the transmission duty cycle δ:

$$\bar{P}_{load} \;=\; P_{MCU,sleep}(1-\delta) \;+\; P_{TX} \cdot \delta \tag{2}$$

With $\delta \approx 0.01\%$ (one transmission every 10 s), $\bar{P}_{load}$ drops below 4 µW, comfortably within the energy budget delivered by the SECE chain. This demonstrates that the combination of SECE extraction circuitry, hybrid PVDF-MoS_2 storage, and ultra-low-power semiconductor peripherals enables *indefinite autonomous operation*, rendering chemical battery replacement logistically and economically obsolete.

Environmental Comparison: Li-ion vs. Supercapacitor In the context of green aeronautics, the lifecycle environmental impact of energy storage components is a primary selection criterion. Lithium-ion batteries require the extraction of critical metals (Li, Co, Ni, Mn) through water-intensive and geopolitically sensitive mining operations; their end-of-life recycling—after typically only 500–2,000 cycles—remains energy-intensive and chemically complex. By contrast, our supercapacitors rely primarily on activated carbon (derivable from biomass), aluminium current collectors, and MoS_2— all recoverable without hazardous wet-chemistry processes. The absence of heavy metals and toxic electrolytes renders the recycling pathway straightforward and low-cost, directly supporting the "zero-lead, zero-toxic" charter of sustainable aviation development.

Criterion	Li-ion Battery	Supercapacitor
Energy Density	High (100–265 Wh/kg). Prolonged storage.	Low (1–10 Wh/kg). Suited to short bursts.
Power Density	Moderate ($\leq$3 000 W/kg)	Very high ($\leq$100 000 W/kg). Ideal for transmission peaks.
Charge Time	Long (minutes to hours)	Extremely fast (seconds to minutes)
Lifespan	Limited (500–2 000 cycles)	Near-unlimited (>1 000 000 cycles)
Energy Efficiency	70–85%	>95% (minimal Joule losses)
ESR	Moderate ($\sim$100 mΩ)	Very low (<10 mΩ); critical for pulse-current SHM loads
Environ. Impact	High. Critical metals (Li, Co, Ni, Mn). Polluting, water-intensive extraction.	Low–moderate. Abundant materials (activated C, Al, graphene).
Safety	Toxic, flammable electrolytes. Thermal runaway risk.	Low toxicity. Fire/explosion risk near zero.
Recycling	Complex, energy-intensive; heavy metal separation still being optimised.	Simple, low-cost; Al and carbon easily recovered.

Table 1. Comparative assessment of Li-ion batteries and supercapacitors across technical, environmental, and safety criteria for embedded autonomous SHM applications.

The integrated architecture described in this section—SECE semiconductor circuitry operating on low-power MOSFET switching, PVDF-MoS_2 hybrid supercapacitors providing high-cycle storage, and duty-cycled autonomous microelectronics keeping average consumption below 5 µW—forms a genuinely self-sufficient, zero-maintenance energy node. This platform was designed not as an isolated laboratory curiosity, but as a deployable solution for environments subjected to continuous structural vibration. The following sections examine its concrete integration into two such environments: **aircraft structural monitoring systems**, where weight reduction and fail-safe autonomy are paramount, and **intelligent railway infrastructures**, where distributed self-powered sensor networks must withstand extreme mechanical shocks and wide thermal excursions while enabling real-time track health monitoring.

2.5. Semiconductors: The Enabling Technology of the Energy Chain

Throughout this report, the term *semiconductor* has appeared repeatedly—in the MOSFET switch of the SECE circuit, in the comparator detecting voltage peaks, and implicitly in the microcontroller of the autonomous sensor node. This section establishes a clear conceptual grounding for this keyword, and maps each semiconductor component to its precise functional role in our harvesting-and-storage architecture.

What Is a Semiconductor?

A semiconductor is a material whose electrical conductivity lies between that of a conductor (e.g., copper, $\sigma \sim 10^7\,\mathrm{S\,m^{-1}}$) and an insulator (e.g., SiO_2, $\sigma \sim 10^{-12}\,\mathrm{S\,m^{-1}}$). The physical origin of this intermediate behaviour lies in the **band gap** E_g: the energy difference between the highest occupied electron states (valence band) and the lowest available empty states (conduction band), as illustrated in Figure 6.

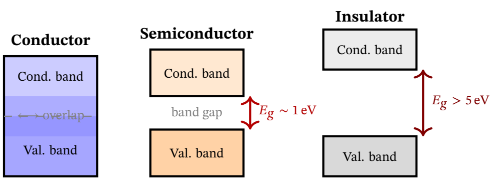

Figure 6. Band diagram comparison of conductor, semiconductor, and insulator. The band gap E_g governs electrical conductivity and determines suitability for active switching devices.

Crucially, the conductivity of a semiconductor can be **controlled externally**—by temperature, light, electric field, or deliberate impurity doping. This controllability is precisely what makes semiconductors the foundation of all active power electronics and signal processing circuits.

The most widely used semiconductor material is silicon (Si, $E_g = 1.12\,\mathrm{eV}$), but wide-bandgap materials such as Silicon Carbide (SiC, $E_g = 3.26\,\mathrm{eV}$) and Gallium Nitride (GaN, $E_g = 3.4\,\mathrm{eV}$) are increasingly relevant to aerospace power electronics due to their higher breakdown voltage, thermal stability, and switching speed.

Semiconductor Devices in Our Energy Harvesting Chain

Each stage of the SECE-based energy management system relies on a specific class of semiconductor device, selected to minimise parasitic losses at the microwatt power levels typical of piezoelectric harvesting. Table 2 maps each functional block to its semiconductor implementation and the critical parameter driving component selection.

Circuit Stage	Device Type	Role in the System	Key Parameter
Peak detection	Analog comparator	Detects the voltage extremum of $V_p(t)$ to trigger the gate drive pulse with sub-microsecond latency.	$I_q < 1\,\mu\mathrm{A}$
Synchronous switch (SECE)	n-MOSFET	Closes the LC resonant tank at the optimal instant; transfers accumulated charge to storage.	$R_{DS,on} < 100\,\mathrm{m\Omega}$
Rectification	Schottky diode	Prevents reverse current flow during LC oscillation; lower V_f than Si diodes minimises conduction loss.	$V_f < 0.2\,\mathrm{V}$
Voltage regulation	Buck-boost converter (MOSFET + diode)	Converts the variable SECE output into a stable supply rail (e.g., 1.8 V or 3.3 V) for the MCU.	$\eta > 90\%$
Sensor & radio	Microcontroller (MCU)	Acquires strain data, executes signal processing, and manages wireless transmission duty-cycle.	$P_{sleep} < 1\,\mu\mathrm{W}$

Table 2. Semiconductor devices at each stage of the piezoelectric energy management chain, with their critical selection parameters.

MoS_2: A Semiconductor at the Heart of Our Storage Material

Beyond classical power electronics, semiconductor physics also governs the performance of our storage material itself. Molybdenum Disulfide (MoS_2) is a **two-dimensional layered semiconductor** with a tunable band gap: in bulk form, $E_g \approx 1.2\,\mathrm{eV}$ (indirect); in monolayer form, $E_g \approx 1.8\,\mathrm{eV}$ (direct). When dispersed as nanosheets within the PVDF matrix, this semiconductor character provides two distinct advantages:

- **Enhanced charge transfer**: The semiconductor band structure of MoS_2 promotes efficient electron hopping between adjacent nanosheet surfaces, reducing the effective internal resistance of the composite electrode and lowering ESR.
- **Band gap reduction of PVDF**: As noted in Section **??**, the insertion of MoS_2 reduces the effective band gap of the PVDF composite from 6.96 eV to 3.48 eV. This dramatically improves the intrinsic piezoelectric-to-electrical charge transfer efficiency at the harvester level, meaning more of the mechanical energy captured by the cantilever is successfully extracted as usable electrical current.

In this sense, the keyword *semiconductor* permeates every layer of our project: from the power MOSFET switching microsecond charge packets in the SECE circuit, to the two-dimensional MoS_2 nanosheet facilitating charge transport within the supercapacitor electrode, and finally to the ultra-low-power microcontroller that decides when to wake, measure, and transmit. Semiconductors are not merely a supporting technology in this system—they are the indispensable enablers of autonomous, lead-free, maintenance-free structural health monitoring.

3. APPLICATIONS AND SYSTEMS INTEGRATION

While the primary catalyst for developing this advanced piezoelectric and hybrid storage architecture is the decarbonization of the aerospace sector, the resulting technology possesses inherent cross-sector versatility. Any large-scale infrastructure subjected to continuous, high-energy mechanical vibrations presents a viable environment for structural energy harvesting. This section details the practical integration of our proposed systems, first examining the primary aerospace use-cases, and subsequently demonstrating how these robust, lead-free solutions can be directly transposed to meet the demanding monitoring requirements of modern railway networks.

3.1. Application to Aircraft Systems

Piezoelectric Harvesters as Power Sources for External Sensors

The integration of piezoelectric technology within aerospace structures operates under two distinct paradigms. The first utilizes the piezoelectric effect strictly as a power generation mechanism to sustain external devices, while the second leverages the material itself as a primary measurement tool. In this configuration, the piezoelectric module functions exclusively as a "vibrational battery" designed to replace traditional copper wiring and chemical power sources. The system harvests kinetic energy from high-fatigue areas (such as engine pylons or landing gear attachments) and routes it through the Synchronous Electric Charge Extraction (SECE) circuitry.

The regulated direct current is then stored in the PVDF-MoS_2 hybrid supercapacitors. This stored power is subsequently discharged to operate independent, external Structural Health Monitoring (SHM) nodes, such as commercial wireless strain gauges or accelerometers, alongside their data transmission modules. To guarantee fail-safe operation and strict energy redundancy, this harvesting network can be paired with other systems (like localized micro-fuel cells), ensuring uninterrupted sensor operation even during flight phases where structural vibrations might temporarily drop below optimal harvesting thresholds.

Self-Sensing Composites (Direct Piezoelectric Sensing)

The second approach marks a shift toward highly integrated "Smart Skins," where the piezoelectric material is not just the power supply, but the sensor itself. Taking advantage of the intrinsic flexibility of thin-film PVDF, the active material can be embedded directly into Carbon-Fiber Reinforced Polymers (CFRP) during the standard composite layup and resin infusion processes.

In this self-sensing architecture, any mechanical stress, bending, or deformation experienced by the composite airframe generates a proportional electrical signal directly from the embedded PVDF layer. This electrical output serves a dual purpose: its voltage amplitude acts as a real-time, measurable data point reflecting structural strain, while the generated charge can be simultaneously captured by the storage unit. By fusing the power generator and the sensor into a single structural element, this paradigm eliminates the need for separate external sensor nodes, drastically reducing system complexity and maximizing the aircraft's mass efficiency.

3.2. Application to Railway Systems

The Need for Autonomous Power Sources:

The transition toward intelligent railway systems has intensified the need for autonomous, distributed power sources to support extensive networks of Structural Health Monitoring (SHM) sensors. Among the various energy harvesting modalities, piezoelectric technology has emerged as a frontrunner due to its high power density, compact from factor, and its ability to convert the significant mechanical kinetic energy of passing trains into electrical energy [1][2]. Unlike solar or wind energy which are intermittent and weather dependent, harversting based on vibrations within the track structure offers a reliable power source directly correlated with the operational frequency of the railway [2].

Different Applications of Piezoelectric Transducers System

Current research categorizes the application of piezoelectric transducers (PZTs) based on their integration points within the railway infrastructure: the rail itself, sleepers, and the ballast [1]. Most designs utilize the indirect piezoelectric effect, where the mechanical stress applied by the train's axle load induces a polarization charge across the material. Recent literature emphasizes the transition from simple cantilever beam configurations to more robust, embedded "bridge-type" or "stack" piezoelectric transducers [1]. For instance, PZT stacks placed beneath the rail foot or integrated into the rail pads capitalize on the high-frequency vibrations and vertical displacement of the rail [2]. These systems are frequency vibrations and vertical displacement of the rail. These systems are effective at capturing the impulsive loading characteristic of high-speed rail, where the passage of multiple axles creates creates a continuous excitation spectrum [2].

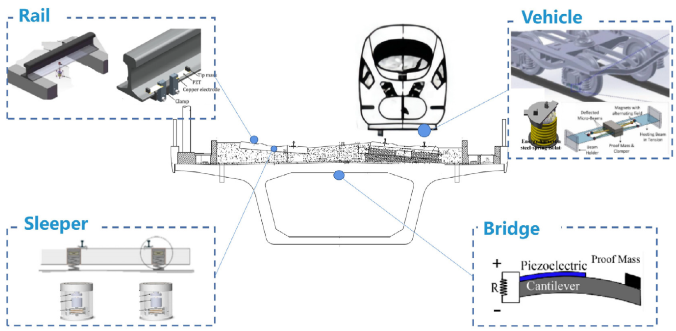

Figure 7. Visual representation of the layout location on railway tracks and trains.

Practical Implementation Issues

The practical implementation of these systems faces significant engineering challenges, primarily related to durability and impedance matching. The railway environment is characterized by extreme mechanical shocks, environmental moisture, and temperature fluctuations, which can lead to the fatigue or depolarization of ceramic piezoelectric materials like Lead Zirconate Titanate (PZT) [1]. To mitigate this, recent advancements have focused on "energy harvesting ties" (sleepers) and the use of protective polymer encapsulation or piezoelectric composites that offer higher flexibility and impact resistance [1]. Furthermore, the electrical output of these transducers is inherently AC and high-impedance, requiring sophisticated rectification and DC-DC conversion circuits to efficiently charge storage units like supercapacitors batteries [1].

Energy Output and Frequency Tuning of Piezoelectric Harvesters

Quantitative comparative analyses in recent studies suggest that while individual piezoelectric units may only produce milliwatts of power, the collective array-based system is sufficient to power wireless sensor nodes for real-time monitoring of track temperature, strain and vibration [2][3]. As the industry moves toward "smart tracks" dynamic response remains the primary frontier. Modern simulations using COMSOL Multiphysics indicate that tuning the resonant frequency of the harvester to the dominant frequencies of the track (often influenced by train speed and axle spacing) is critical to maximizing the energy yield and ensuring the long-term viability of self-powered railway monitoring ecosystems [2][3].

4. BUSINESS PLAN AND INDUSTRIAL STRATEGY

As a specialized R&D bureau, our objective is to transition from a laboratory-scale prototype to a viable industrial product. This section outlines the market landscape, the economic justification for our technology, and the industrial model required to maintain a competitive edge.

4.1. Market Analysis and Targeted Consumers

The aviation industry's is on transition toward More Electric Aircraft. This transformation presents an opportunity for localized, decentralized energy harvesting. Currently, structural health monitoring (SHM) sensors and autonomous IoT nodes require some heavy wiring networks. By positioning this technology as a Tier 2 solution for OEMs (e.g., Airbus, Dassault) and Tier 1 integrators (e.g., Safran), we target an industry critical point: the weight and maintenance of copper wiring.

The primary value driver for the consumer is the reduction of the operating empty weight. Replacing wired power distribution with autonomous, vibration-driven piezoelectric nodes translates to direct fuel savings and lower CO_2 emissions over the aircraft's lifecycle. Furthermore, by prioritizing flexible, lead-free materials (such as PVDF composites) over traditional, brittle piezoceramics (like PZT), and pairing them with high-cycle-life supercapacitors, the system offers an install-and-forget, zero-maintenance guarantee that chemical batteries cannot match under aerospace vibration spectrums.

4.2. R&D Investment and Economic Strategy

Transitioning this technology into a market-ready product requires a lot of investment in engineering and materials science. Given the complexities of electromechanical coupling, the initial R&D budget must be heavily allocated toward establishing a robust digital twin environment. High-fidelity multiphysics simulations are essential to optimize the PVDF-MoS_2 hybrid films and predict energy yields without relying on costly, iterative physical prototyping.

To balance these upfront intellectual property investments, an aggressive and distributed manufacturing strategy is proposed. High fixed costs and labor rates in Europe make local mass-manufacturing of standard storage cells economically inefficient. Therefore, while R&D, system integration, and the tuning of the power management circuits (SECE) will remain in-house to protect our core technology, the mass production of the supercapacitor cells will be offshored. Leveraging specialized subcontractors in regions with optimized energy costs will drive down the unit price, making the system highly competitive.

4.3. Production Scalability and Logistics

To meet the demands of an aerospace assembly line, our production model must be scalable. Since the production is high-volume and demands high-reliability, the production must separate "smart" integration from "passive" manufacturing. The engineering team will focus entirely on the precision tuning of the piezoelectric harvesters to specific aircraft resonance frequencies and the assembly of the smart control units.

Additionally, this strategy seek long-term supply chain resilience. By utilizing carbon-based supercapacitors and easily obtainable polymer substrates, the business model mitigates the geopolitical and logistical risks associated with rare-earth metals and battery minerals like lithium or cobalt. This independence not only stabilizes production costs but also strengthens the environmental compliance of the final product.

4.4. Market Size, Growth, and IoT Replacement Potential

The economic viability of our technology is supported by the rapid expansion of the global piezoelectric market. The broader piezoelectric devices market was valued at approximately $35.59 billion in 2024 and is projected to reach $55.49 billion by 2030 (a CAGR of 7.7%). More specifically, the *piezoelectric energy harvesting* sector—our direct operational niche—is experiencing even more aggressive growth. Valued at $1.8 billion in 2025, it is forecast to reach $4.7 billion by 2034 with a Compound Annual Growth Rate (CAGR) of 11.2%. This accelerated growth is heavily driven by the aerospace, automotive, and industrial sectors actively seeking to eliminate wiring weight and battery maintenance.

The true scale of our financial opportunity lies in the replacement of conventional chemical micro-batteries for autonomous systems. Global IoT (Internet of Things) device deployments are forecast to exceed 29 billion active connections by 2030. Currently, the vast majority of these wireless structural and environmental sensors rely on chemical batteries that have a limited lifespan and require costly manual replacement.

If our piezoelectric PVDF-MoS_2 and supercapacitor system captures even 1% of this micro-consumption battery replacement market for industrial and aerospace sensors, that represents 290 million units deployed. Assuming an average B2B unit price of $10 to $15 for a fully integrated smart node, the addressable revenue purely in battery-replacement scenarios reaches $2.9 billion to $4.3 billion. We are not just selling power generation; we are selling the permanent elimination of maintenance logistics.

4.5. Smart Grids and Energy Storage Synergy

Looking toward the future, combining piezoelectric harvesting with advanced storage (supercapacitors) positions us perfectly for the localized "Smart Grid" concept within an aircraft or factory. Instead of a central power hub pushing electricity through miles of copper to thousands of sensors, our devices create decentralized, self-sufficient micro-grids. The harvester continuously generates power during mechanical stress (e.g., flight turbulence, engine vibration), and the storage cell banks it to power data transmission bursts. By self-powering these micro-grids, we drastically reduce the parasitic load on the aircraft's main generators, offering another layer of measurable financial and environmental value to our clients.

5. PERSPECTIVES: THE FUTURE OF PIEZOELECTRICITY

As the aerospace industry moves towards "Zero-Emission" targets, the role of energy harvesting must evolve beyond simple secondary power sources. Our R&D bureau anticipates a shift from discrete components to fully integrated, bio-compatible, and intelligent energy systems.

5.1. Technological Evolution: Bio-sourced and Sustainable Materials

The current transition from PZT to lead-free ceramics is only the first step toward true sustainability.

- **Biodegradability**: Future research aims to exploit the natural piezoelectric properties of wood (cellulose) and collagen. These bio-polymers offer a path toward sensors that are not only lead-free but fully biodegradable, reducing the environmental footprint of decommissioned aircraft.
- **Life-Cycle Circularity**: Unlike current synthetic polymers, bio-sourced piezoelectric materials can be processed at much lower temperatures, further reducing the carbon intensity of the manufacturing phase.

5.2. Miniaturization and Functional Fluids: The "Smart Dust" Concept

The integration of piezoelectricity is moving toward the microscopic scale, enabling monitoring in environments previously considered inaccessible.

- **SAF Integration**: We envision the deployment of "Smart Dust" — microscopic piezoelectric transducers dispersed directly within *Sustainable Aviation Fuels* (SAF).
- **Real-time Fluid Analysis**: These micro-sensors would harvest energy from the turbulent flow within fuel lines to power real-time chemical analysis, ensuring fuel quality and combustion efficiency without external wiring or bulky battery packs.

5.3. 2040 Roadmap: Towards Auto-Capacitive Airframes

The ultimate goal of our R&D strategy is the total fusion of power generation, storage, and airframe structure.

- **Structural Energy Storage**: By 2040, the aircraft fuselage will likely act as a giant hybrid supercapacitor. By integrating PVDF-MoS2 architectures directly into the carbon-fiber layup, the airframe will simultaneously harvest vibration energy and store it within its own mass.

- **Elimination of Buffer Batteries**: This "auto-capacitive" structure will render traditional buffer batteries obsolete. The aircraft becomes a self-sustaining organism where every vibration, from takeoff to landing, contributes to a distributed energy network, maximizing OEW (Operating Empty Weight) efficiency and reliability.

6. CONCLUSION

This study demonstrates that the integration of a PVDF-MoS_2 piezoelectric harvesting chain, coupled with SECE conditioning circuitry and hybrid solid-state supercapacitor storage, constitutes a technically viable and ecologically coherent alternative to conventional wired or battery-powered SHM systems.

The substitution of lead-based PZT by aluminium nitride and PVDF composites addresses both the regulatory trajectory imposed by the RoHS directive and the structural brittleness inherent to ceramic transducers. Meanwhile, the SECE interface delivers a conversion efficiency improvement exceeding 400% over standard diode bridges, enabling indefinite autonomous operation at average load levels below 5 µW.

From an industrial standpoint, the proposed R&D model—combining in-house system integration with offshored cell manufacturing—offers a credible path to cost competitiveness within a piezoelectric energy harvesting market projected to reach $4.7 billion by 2034. By eliminating toxic materials, mechanical connectors, and scheduled battery replacements in a single architecture, this platform positions itself not merely as an engineering improvement, but as a structural enabler of next-generation autonomous, zero-maintenance aircraft.

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